Bow Valley College

Course Outline

TECH9210

Switching, Routing, and Wireless Essentials
Winter 2023 - Spring 2025

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TECH9210 Switching, Routing, and Wireless Essentials

COURSE DESCRIPTION

The Switching, Routing, and Wireless Essentials course is focused on switching technologies and router operations that support small-to-medium business networks. It includes wireless local area networks (WLANs) and security concepts as well as key switching and routing concepts. Learners perform basic network configuration and troubleshooting, identify, and mitigate LAN security threats, and configure and secure a basic WLAN.

REQUISITES	None
EQUIVALENTS	None
CREDITS	0
HOURS	128
ELIGIBLE FOR PLAR	No
ZERO TEXTBOOK COST	

COURSE LEARNING OUTCOMES

Bow Valley College is committed to ensuring our graduates can demonstrate their abilities in key areas that will make them effective citizens and encourage their development as lifelong learners. In addition to the discipline-specific skills that learners acquire in their programs, the College has identified ten learning outcomes.

College-Wide Outcomes:

1. Communication
2. Thinking Skills
3. Numeracy and Financial Literacy
4. Working with Others
5. Digital Literacy
6. Positive Attitudes and Behaviours
7. Continuous Learning
8. Health and Wellness Awareness
9. Citizenship and Intercultural Competence
10. Environmental Sustainability

#	COURSE LEARNING OUTCOME(S)	COLLEGE WIDE OUTCOMES SUPPORTED
1	Configure Virtual Local Area Network (VLANs) and Inter-VLAN routing, and apply security best practices.	1, 2, 5
2	Troubleshoot inter-VLAN routing on Layer 3 devices.	1, 2, 4, 5
3	Configure redundancy on a switched network using Spanning Tree Protocol (STP) and EtherChannel.	1, 2, 5
4	Explain how to support available and reliable networks using dynamic addressing and first-hop redundancy protocols.	1, 2, 5
5	Configure dynamic address allocation in IPv6 networks.	2, 5
6	Configure Wide Area Networks (WLANs) using a Wireless LAN Controller (WLC) and Layer 2 (L2) security best practices.	1, 2, 5
7	Configure switch security to mitigate Local Area Network (LAN) attacks.	1, 2, 5
8	Configure IPv4 and IPv6 static routing on routers.	1, 2, 5

COURSE MODULES AND SCHEDULE

**Course schedule subject to change, depending on delivery mode and term of study. For exact dates, please consult the Course Offering Information in Brightspace.*

WEEK/HOURS	MODULES
Week 1	Basic device configuration, switching concepts, Virtual Area Networks (VLANs), and Inter-VLAN routing
Week 2	Spanning Tree Protocol (STP) concepts, EtherChannel, Dynamic Host Configuration Protocol v4 (DHCPv4), Stateless Address Autoconfiguration (SLAAC), and Dynamic Host Configuration Protocol v6 (DHCPv6)
Week 3	First Hop Redundancy Protocol (FHRP) concepts, Local Area Network (LAN) security concepts, switch security configuration, and Wireless LAN (WLAN) concepts
Week 4	WLAN configuration, routing concepts, IP static routing, troubleshoot static and default routes, and final assessments

ASSESSMENT

COURSE LEARNING OUTCOME(S)	ASSESSMENT	WEIGHT
1, 2, 3, 4, 5, 6, 7, 8	Practical application	50%
1, 2, 3, 4, 5, 6, 7, 8	Final exam	50%

Important: For details on each assignment and exam, please see the Course Offering Information.

PERFORMANCE STANDARDS

A grade of P is required to pass this course. Learners may be required to pass this course in order to progress in the program or to meet specific program completion requirements.

To receive a course overall pass, the learner must pass all assessments.

Please consult with the program area or the Bow Valley College website for further details.

GRADING SCHEME

Symbol	Description	Grade Point Value
P	Pass	N/A
F	Fail	N/A
BD	Basic/Developing (Foundational programming courses only)	N/A
CM	Complete	N/A
NCM	Not Complete	N/A

REQUIRED LEARNING RESOURCES

Computer requirements:

Standards	**Hardware**	***Software***
Processor	i7	
RAM (memory)	16 GB RAM or greater	Windows 10 Pro/education 64-bit
Hard drive & storage	512 GB SSD or greater	
Video card	On-board integrated	(MacOS is not supported)
Screen size	15" or greater	

Additional learning resources may be found in the Course Offering Information or in Brightspace.

ADDITIONAL INFORMATION

Additional information may be found in the Course Offering Information or in Brightspace.

ACADEMIC ACCOMMODATIONS

Learners with a disability (learning, physical, and/or mental health) may qualify for academic and exam accommodations. For more information, or to apply for accommodations, learners should make an appointment with Accessibility Services in the Learner Success Services (LSS) Department. Accessibility Services can also assist learners who may be struggling with learning but do not have a formal diagnosis. To make an appointment visit LSS on the first floor of the south campus or call 403-410-1440. It is the learner's responsibility to contact Accessibility Services and request academic accommodations. For more information, please visit our website at http://www.bowvalleycollege.ca/accessibility.

INSTITUTIONAL POLICIES

Bow Valley College is committed to the highest standards of academic integrity and honesty. Learners are urged to become familiar with and uphold the following policies: Academic Integrity (500-1-7), Learner Code of Conduct, Procedures and Guidelines (500-1-1), Learner Appeals (500-1-12), Attendance (500-1-10), Grading (500-1-6), Academic Continuance and Graduation (500-1-5), and Electronic Communications (300-2-13). Audio or video recording of lectures, labs, seminars, or any other teaching and learning environment by learners is allowed only with consent of the instructor as part of an approved accommodation plan. Recorded material is to be used solely for personal study and is not being used or distributed without prior written consent from the instructor.

Turnitin:

Students may be required to submit their course work to Turnitin, a third-party service provider engaged by BVC. Turnitin identifies plagiarism by checking databases of electronic books and articles, archived webpages, and previously submitted student papers. Students acknowledge that any course work or essays submitted to Turnitin will be included as source documents in the Turnitin.com reference database, where it will be used solely to detect plagiarism. The terms that apply to a student's use of Turnitin are described on Turnitin.com.

Online Exam Proctoring:

Examinations for this course may require proctoring through an online proctoring service. Online proctoring enables online exam taking within a controlled and monitored environment, thereby enhancing academic integrity. Online proctoring may occur through a variety of methods, including but not limited to:

a. live online proctoring where a remote invigilator authenticates identity and observes completion of an exam using specialized software and recordings;
b. automated proctoring where the exam session is recorded and AI (artificial intelligence) analyzed;
c. browser lockdown that limits access to other applications, websites, copying, printing, screen capture and other functions; or
d. a combination of both live/automated proctoring and browser lockdown.

Course instructors will review recordings, analyses, and data obtained through online proctoring for academic integrity infractions. It is the student's responsibility to meet the technical, software, location, and identity verification requirements necessary to enable online proctoring.

Further details of these policies are available in the Academic Calendar and on the Bow Valley College website, bowvalleycollege.ca.

Learners are encouraged to keep a copy of this course outline for future reference.

Collection of Personal Information:

This course, including your image and voice, may be recorded and made available to you and other students taking the course section. By attending the class(es) online or in person, you consent to the collection of your personal information. If you do not wish to be recorded, please contact your instructor before starting the course/class to discuss alternative arrangements.

You may use the recordings only for educational purposes and you must not copy, share, or use the recordings for any other purpose without the instructor's express permission.

Your personal information is collected in accordance with section 33(c) of the Freedom of Information and Protection of Privacy Act (Alberta) to deliver academic programming, support learner flexibility, promote universal design for learning principles, and for purposes consistent with the course activities and outcomes. If you have any questions about the collection, disclosure, use, or protection of this information, please contact the College's Access and Privacy Officer at foip@bowvalleycollege.ca.